EXPERIMENT No.: 6

Student Name: Sujal Srivastava **UID:** 23BCS11842
Branch: BE-CSE **Section/Group:** KRG-3-B
Semester: 6th **DOP:24th FEB** 2026
Subject Name: SYSTEM DESIGN **Subject Code:** 23CSH-314

1. Introduction

Video streaming platforms such as YouTube require efficient systems to upload, process, store, and deliver videos to millions of users. This project demonstrates a **scalable video streaming architecture** that uses **Amazon S3, CDN, API Gateway, Load Balancer, and video transcoding services** to manage video uploads and deliver optimized video content to clients.
The system ensures:

- Efficient video upload
- Automated video processing
- Storage of video metadata
- Delivery of optimized video chunks through CDN
- Support for multiple video resolutions

---

2. System Overview

The architecture consists of several components that work together to manage video uploads, processing, and streaming.
Main components include:

- Client
- API Gateway
- Load Balancer
- Amazon S3 Storage
- Video Metadata Database
- Chunker Service
- Transcoding Services
- Content Delivery Network (CDN)

Each component performs a specific function in the video streaming pipeline.

---

3. Client Interaction

The process begins when a **client uploads a video** to the platform.

Steps performed by the client:

1. The client sends a request to the backend system through the **API Gateway**.
2. The system generates **pre-signed URLs** that allow secure uploads to Amazon S3.
3. The client uploads the video directly to **Amazon S3 using multipart upload**.
4. The client later downloads video content through the **CDN**, which delivers cached video chunks efficiently.

Direct upload to S3 reduces server load and improves scalability.

---

4. API Gateway and Load Balancer

The **API Gateway** acts as the entry point for all client requests.

Responsibilities:

- Handle authentication and request validation
- Generate **pre-signed upload URLs**
- Forward requests to backend services

The **Load Balancer** distributes requests across multiple backend servers to ensure:

- High availability
- Fault tolerance
- Efficient resource utilization

---

5. Amazon S3 Storage

Amazon S3 serves as the **primary storage system for videos**.

Functions:

- Store original uploaded video files
- Store processed video chunks
- Store different resolution versions of videos

Advantages of S3:

- High durability
- Scalability
- Secure storage
- Integration with event notifications

When a video is uploaded to S3, it triggers **S3 Notification events**.

---

6. Video Metadata Database

The system stores video information in a database.

Metadata stored includes:

Field	Description
Video ID	Unique identifier
Name	Video title
Description	Video details
Status	Processing stage
Chunks	Number of video chunks
Full S3 URL	Video storage location

Video status may include:

- pending
- uploaded
- chunking
- complete

This database helps track the processing state of each uploaded video.

---

7. S3 Notification System

When a video upload is completed, **Amazon S3 triggers a notification event**.
This event activates backend services responsible for:

- Video chunking
- Video transcoding
- Processing workflow

This event-driven architecture enables **automatic video processing** without manual intervention.

---

8. Video Chunking Service

The **Chunker Service** divides the uploaded video into smaller segments called **chunks**.
Purpose of chunking:

- Enable adaptive streaming
- Improve playback performance
- Reduce buffering

The chunked files are stored back in **Amazon S3**.
These chunks allow the system to stream only the required segments during playback.

---

9. Video Transcoding

The **Transcoding Service** converts videos into multiple resolutions.
Supported formats include:

- 240p
- 720p
- 1080p
- 4K

Advantages of transcoding:

- Supports different devices and internet speeds
- Enables adaptive bitrate streaming
- Improves user experience

Each resolution is stored separately in S3.

---

10. Content Delivery Network (CDN)

A **CDN (Content Delivery Network)** is used to distribute video chunks efficiently.

Functions:

- Cache popular video chunks
- Deliver content from edge locations
- Reduce latency
- Improve playback speed

When a client requests a video:

- The CDN checks if the chunk exists in cache.
- If available, it serves the chunk directly.
- Otherwise, it fetches it from S3 and caches it.

---

11. Adaptive Video Streaming

The system supports **adaptive streaming**, meaning the video quality automatically adjusts based on the user's internet speed.

Example:

- Slow network → 240p video
- Medium network → 720p video
- High speed → 1080p or 4K

This ensures smooth playback with minimal buffering.

---

12. Advantages of the Architecture

1. Scalability

Using S3 and CDN allows the system to handle millions of video requests.

2. High Availability

Load balancers ensure the system remains operational even if some servers fail.

3. Cost Efficiency

Direct upload to S3 reduces backend server costs.

4. Faster Streaming

CDN caching reduces latency and improves video delivery.

5. Automated Processing

S3 event notifications automate video processing workflows.
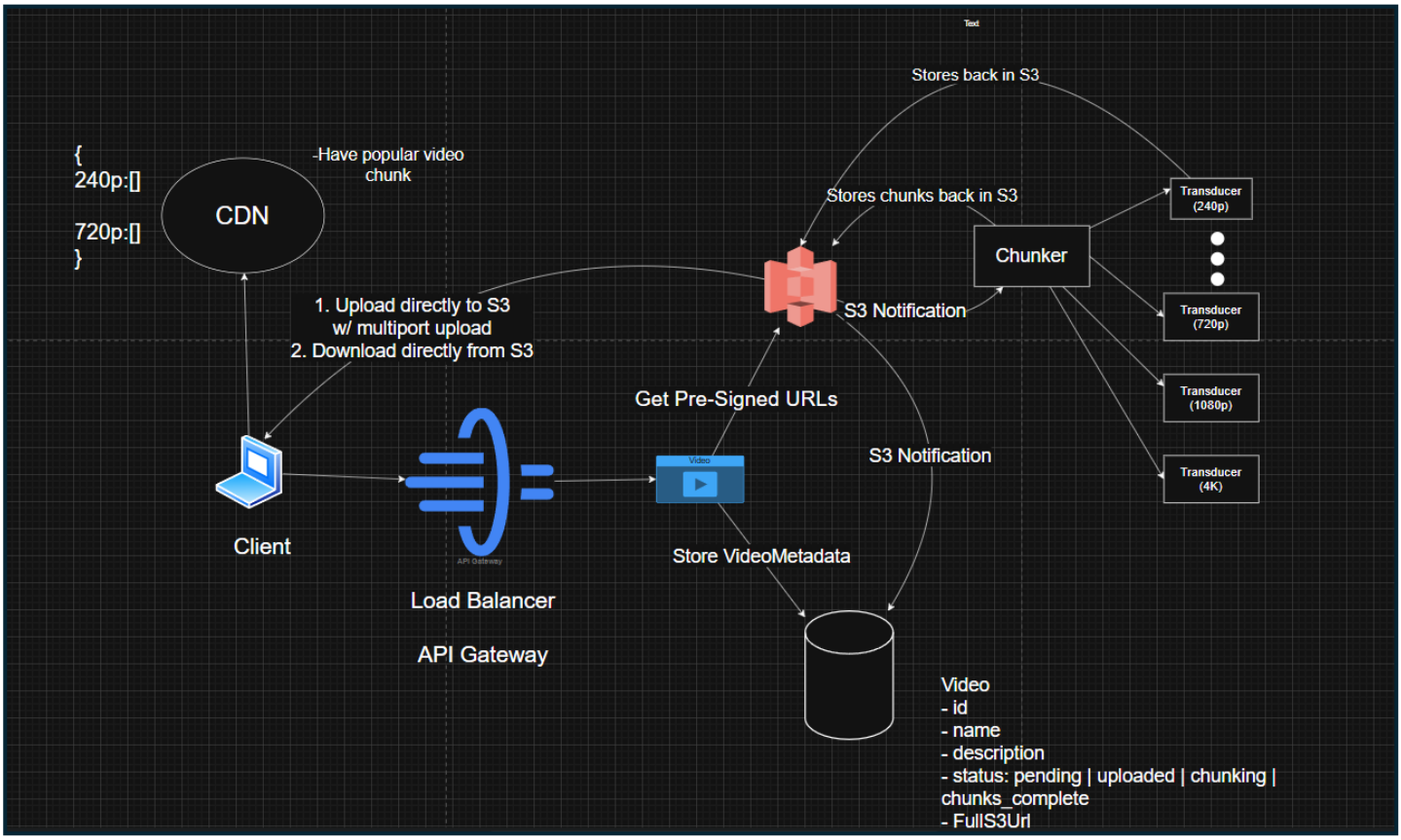
DIAGRAM: